

KENYA NATIONAL EXAMINATIONS COUNCIL

KCPE 2007

SCIENCE

READ THESE INSTRUCTIONS CAREFULLY

Time:1 hour 40 minutes

1. You have been given this question booklet and a separate answer sheet. The question booklet contains 50 question
2. Do any necessary rough work in this booklet.
3. When you have chosen your answer, mark it on the **ANSWER SHEET**, not in this question booklet.
4. Use only an ordinary pencil.
5. Make sure that you have written on the answer sheet:

YOUR INDEX NUMBER

YOUR NAME

NAME OF YOUR SHOOOL

6. By drawing a dark line inside the correct numbered boxes mark your full Index Number(i.e School code number and the three-figured Candidate's number)in the grid near the top of the answer sheet.
7. Do not make any marks outside the boxes.
8. Keep the sheet as clean as possible and do not fold it.
9. For each of the question 1-50 four answers are given. The answers are lettered A,B,C and D. In each case only ONE of the four answers is correct. Choose the correct answer.
10. On the answer sheet the correct answer is to be shown by drawing a dark line inside the box in which the letter have chosen is written.

Example

In the Question booklet.

4. Which one of the following components of blood is involved in the clotting after injury?
- A. Plasma
 - B. White blood cells
 - C. Red blood cells
 - D. Platelets

The correct answer is D

1 [A] [B] [C] [D] 14. [A] [B] [C] [D] 24. [A] [B] [C] [D] 34. [A] [B] [C] [D]

11. Your dark line **MUST** be within the box.

12. For each question **ONLY ONE** box is to be marked in each set of four boxes.

This question paper consists of 7 printed pages and 1 black page.

The Kenya National Examination Council, 2007.

1. Which one of the following is **NOT CORRECTLY** matched with its function?

Part	Function
A. Oesophagus	passage of food.
B. Stomach	mixes food with enzymes.
C. Small intestine	completes digestion of food.
D. Large intestine	absorbs digested food.

2. Which of the following is **NOT A CORRECT** statement about reproduction in human beings?

- A. Fertilization takes place in the uterus.
- B. The zygote gets implanted in the wall of uterus.
- C. The blood of the mother and the foetus not to mix.
- D. Menstruation stops during Pregnancy.

3. The blood vessels through which deoxygenated bloods flows to the heart are the

- A. Aorta
- B. Pulmonary artery
- C. Vena cava
- D. Pulmonary vein

4. Which one of the following changes takes place during adolescence in boys only?

- A. Fast body growth.
- B. Growth of pubic hair.
- C. Chest and shoulder become broader.
- D. Releasing sex cell.

5. Which one of the following statements is **TRUE** about HIV and AIDS?

- A. All HIV positive people have AIDS.
- B. During the window stage a person may test negative.
- C. It can be transmitted by sharing utensils.
- D. It is not spread during asymptomatic phase.

6. Which one of the following is **NOT** communicable disease?

- A. Whooping cough.
- B. Tetanus.
- C. Tuberculosis.

- D. Measles.
7. A child suffering from kwashiorkor should be given diet rich in?
- A. Carbohydrate.
 - B. Fats.
 - C. Vitamins.
 - D. Proteins.
8. Wearing proactive clothing while walking in stagnant water can prevent one from contracting
- A. Bilharzias.
 - B. Typhoid.
 - C. Typhoid and cholera.
 - D. Bilharzias and cholera.
9. Which one of the following statements is correct according to immunization schedule of infants?
- A. The first dose of polio is given at six weeks.
 - B. DPT vaccine is given at birth and at 10 weeks.
 - C. DPT and measles vaccine is given at 14 weeks.
 - D. Polio and BCG vaccine is given at birth.
10. Planets appears bright because they
- A. Reflect light.
 - B. Produce light.
 - C. All twinkle.
 - D. Are large.
11. A pupil gave a following reasons as to why dairy cattle are given commercial feeds
- i) *To increase milk production.*
 - ii) *Given only to cattle under zero grazing.*
 - iii) *To supplement other feeds.*
 - iv) *Only when fodder crops and pasture are not available.*

Which two reasons are **CORRECT**?

- A. (i) and (iii).
 - B. (ii) and (iv).
 - C. (i) and (iv).
 - D. (ii) and (iii).
12. The diagram below represent a set up that is used to left a load.

DIAGRAM

To investigate the force required to lift the load a spring balance must be at

- A. **P**
- B. **Q**
- C. **R**
- D. **S**

13. The following are some methods of preserving foods?

- i) *Smoking*
- ii) *Drying*
- iii) *Freezing*
- iv) *Salting*
- v) *Canning*

Which methods are traditional?

- A. (ii) (iv) and (v).
- B. (ii) (iii) and (v).
- C. (i) (iii) and (iv).
- D. (i) (ii) and (iv).

14. Which one of the following pairs of machines belong to the same group as ladder?

- A. Claw hammer and bottle opener.
- B. Spade and fishing rode.
- C. Staircase and a road winding up a hill.
- D. Handcart and wheel barrow.

15. The reason why an ordinary *jiko* without clay lining is not recommended for use is because

- A. It pollutes the environment.
- B. Of the large amounts of charcoal used.
- C. It takes a long time when cooking.
- D. It gets hot to hold.

16. The following are liquids added together in a glass bottle labeled **P**, **Q**, **R** and **S** by some pupils.

- P** –kerosene and cooking oil.
- Q** –water and cooking oil.
- R** –milk and water.
- D** –milk and kerosene.

Which glass bottle contains liquids that cannot be separated by decanting?

- A. **P**.
- B. **Q**.

C. R.

D. S.

17. Which one of the following plants is green and non- flowering?

A. Mould.

B. Onion.

C. Sisal

D. Moss.

18. Which one of the following consists of pairs of storage past?

A. Rats and aphids.

B. White ants and rats.

C. Aphids and cutworms.

D. White ants and cutworms.

19. Which one of the following is the possible food chain that could be found in grassland inhabited by hawks, frogs, snakes, mice and grasshoppers?

A. Grass-> grasshoppers-> frogs->snakes.

B. Grass-> grasshoppers-> frogs->mice.

C. Grass-> snakes->mice-> hawks.

D. Grass-> grasshoppers->hawks->snakes.

20. Which one of the following is **TRUE** for the both chickens and frogs? Both

A. Has a constant body temperature.

B. Have scale.

C. Breath by lungs.

D. Feed on insects only.

21. Which one of the following is **NOT** a rotational method of grazing?

A. Paddockking.

B. Herding.

C. Tethering.

D. Strip.

22. Birds which feed on nectars have

A. Short pointed beaks.

B. Strong curved beaks.

C. Long slender pointed beak.

D. Long slender curved beak.

23. In which of the following practices is water re-used?

A. Using dirty water from the kitchen for watering plants.

B. Collecting rain water from the roof for domestic use.

C. Getting clean water from dirty water by boiling and cooling the steam.

D. Storing water in the dams to be used in the irrigation.

24. The texture of the soil type depends on the
- A. Amount of water in the soil.
 - B. Organic matter in the soil.
 - C. Amount of the mineral particle in the soil.
 - D. Size of the soil particle.
25. The component that make up 0.97% of air is used in
- A. Rusting.
 - B. Preservation of soft drinks.
 - C. Electric bulbs.
 - D. Making proteins.
26. Which one of the following pairs of processes is as a results of increase in temperature?
- A. Melting and evaporation.
 - B. Condensation and freezing.
 - C. Condensation and evaporation.
 - D. Melting and freezing.
27. The diagram below shows a set up that was used to demonstrate a certain property of matter.

DIAGRAM

The coloured water rises up the straw because

- A. Liquids expand when heated.
 - B. Liquids occupy space.
 - C. Air occupies space.
 - D. Air expands when heated.
28. A *sufuria* made up of steel weighing 1000g floats on water while a nail made of iron weighing 50g sinks.
- This is because of difference in
- A. Type of material.
 - B. Shape.
 - C. Volume.
 - D. Mass.
29. Which one of the following materials will **NOT** be required in an experiment to show that pressure in liquid increase with depth?

- A. Tin can.
- B. Water.
- C. Nail.
- D. Collecting jar.

30. The diagram below represents a circuit that can be used to make a temporary magnet

The magnet transformation that occurs at the section marked **P** is

- A. Electrical -> magnetic.
- B. Chemical->electrical.
- C. Heat->electricity.
- D. Chemical->magnetic.

31. The following are forms of energy:

- (i) *Heat*
- (ii) *Electricity*
- (iii) *Light*
- (iv) *Sound*

The **two** forms of energy that **DO NOT** require a medium of transmission are

- A. (ii) and (iv).
- B. (i) and (iii).
- C. (i) and (ii).
- D. (iii) and (iv).

32. The following practices help to conserve energy:

- i) *Using improved firewood jikos.*
- ii) *Switching off light that use hydroelectric power when not needed.*
- iii) *Using public transport whenever possible.*
- iv) *Using biogas for cooking.*

Which **two** practices conserve non- renewable

- A. (i) and (ii).
- B. (ii) and (iiii).
- C. (ii) and (iv).
- D. (iii) and (iv).

33. Which one of the following is a recommended practice to avoid being struck by lightning?

- A. Sheltering under a tree.
- B. Using an umbrella when in an open field.
- C. Sheltering while leaning on a wall.
- D. Wearing rubber shoes with thick soles.

34. Which one of the following groups consists only foodstuffs that are used for body building and repair?

- A. Rice, oranges, meat.

- B. Wheat, fish, maize.
 - C. Beans, potatoes groundnuts.
 - D. Eggs, peas, milk.
35. The function of fibre in the human diet is to
- A. Provide the body with nutrients.
 - B. Transport digested food.
 - C. Prevent constipation.
 - D. Help in digestion of food.
36. Which one of the following statements is **NOT TRUE** about the force that oppose movements the force
- A. Can be reduced by streamlining.
 - B. Makes the load heavier.
 - C. Increase efforts required to move the load.
 - D. Produces heat.
37. A farmer noticed small and shallow channels forming in the soil in the garden due to running water. This type of soil erosion is called
- A. Splash.
 - B. Rill.
 - C. Sheet.
 - D. Gully.
38. Which one of the following plants is cereal?
- A. Millet.
 - B. Coffee.
 - C. Sunflower.
 - D. Peas.
39. Which one of the following consists of a pair of plants that **CANNOT** make their own food?
- A. Fern and pine.
 - B. Toadstool and moss.
 - C. Puffballs and mould.
 - D. Cedar and yeast.
40. Which one of the following pairs of parasites can be controlled by spraying?
- A. Ticks and fleas.
 - B. Liver flukes and tsetse flies.
 - C. Fleas and liver flukes.
 - D. Ticks and tape worms.
41. Which one of the following source of electricity pollutes the environments
- A. Geothermal.
 - B. Wind-driven generators.
 - C. Hydro-electric generator.

- D. Petrol –driven generators.
42. Which one of the following is a natural way of lighting a house? Using a
- Candle.
 - Torch.
 - Hurricane lamp.
 - Transparent roof.
43. Which one of the following components is found in all environments?
- Plants.
 - Soil.
 - Air
 - Water.
44. Which one of the following groups of methods can be used to separate solid mixtures only
- Sieving, picking, winnowing.
 - Picking, use of magnet, decanting.
 - Winnowing, sieving, filtration.
 - Use of magnet, filtration, decanting.
45. Which one of the following does not function the same way as bicycle dynamo when used to produce electricity?
- Hydro-electric generator.
 - Car battery.
 - Diesel generator.
 - Wind driven turbines.
46. The diagram below represents a simple machine.

DIAGRAM

Which of the position labelled **P**, **Q** and **R** would represent the load effort and fulcrum when the machine is use?

	Load	Effort	Fulcrum
A.	P	Q	R
B.	R	P	Q
C.	Q	P	R
D.	R	Q	P

47. The diagram below represents a block of pushed along a surface towards point **X**.

Which one of the arrows represents the direction of the force of friction?

- J.

- B. K.
- C. L.
- D. M.

48. Which one of the following consists of only materials that are **ALL** non –magnetic?

- A. Aluminum, foil, shilling coin.
- B. Iron, nail, staple pins.
- C. Metal scissors, steel wool.
- D. Pieces of glass, sewing needle.

49. The reason why coloured water is used in construction of liquid thermometer is to

- A. Make it visible.
- B. Make it expand faster.
- C. Make it absorb more heat.
- D. Increase its density.

50. Which one of the following is the largest planet?

- A. Mercury.
- B. Saturn.
- C. Jupiter.
- D. Earth.